

Multivariate Statistical Analysis (MA304)

Group Project Report

A Study on Precise Matching between Securities Investment Strategies and Client Trading Profiles

Group Members

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Abstract

With the rapid development of quantitative investment services, financial institutions need effective methods to recommend suitable trading strategies to clients with different investment styles, risk preferences, and transaction behaviors. This project studies the problem of matching stock quantitative strategies with client trading profiles. The main objective is to construct an interpretable and data-driven framework that can identify similarities between clients and strategies, and further recommend strategies that not only resemble the client’s historical trading style but also achieve better risk-return performance.

Based on strategy transaction records and simulated client account data, this report builds a multivariate feature system from three perspectives: trading behavior, portfolio structure, and risk-return characteristics. The final dataset contains 20 quantitative strategies, 3 real simulated client accounts, and 25 additional simulated clients generated from observed client features and business assumptions, forming 48 analytical entities in total. Eighteen variables are extracted, including trading frequency, average holding period, turnover rate, portfolio concentration, win rate, maximum drawdown, return volatility, and approximate Sharpe ratio.

Several multivariate statistical methods are used in this study. Principal component analysis is first applied to explore the major dimensions of variation in the feature space. K-means clustering, hierarchical clustering, and Gaussian mixture models are then used to examine whether clients and strategies form stable groups. Fisher’s linear discriminant analysis is further used to visualize the separation among clusters. Finally, a two-stage matching model is constructed by combining style similarity and performance improvement.

The empirical results show that three clusters provide a reasonable and interpretable grouping structure. The selected K-means model achieves a silhouette coefficient of 0.4203. The adjusted Rand index between K-means and hierarchical clustering is 0.8438, while that between K-means and the Gaussian mixture model is 0.9699, indicating that the clustering structure is relatively stable across methods. For the three real simulated clients, the top recommended strategies all improve both return and maximum drawdown compared with the clients’ approximate historical performance. These findings suggest that the proposed framework can serve as a feasible preliminary solution for client-strategy matching in a securities investment context.

1 Introduction

1.1 Background of the Problem

In recent years, quantitative investment strategies have become increasingly important in securities markets. Different strategies may differ significantly in stock selection rules, trading frequency, holding period, portfolio concentration, volatility, and drawdown control. For example, some strategies focus on large-cap enhancement, some emphasize growth stocks or sector rotation, while others rely on momentum signals or balanced allocation. As a result, a single strategy is unlikely to be suitable for all investors.

At the same time, clients of securities institutions also have heterogeneous trading behaviors and risk preferences. Some clients trade infrequently and hold concentrated portfolios, while others trade actively and diversify their positions across many stocks or industries. After clients open accounts with securities institutions, their historical transaction records naturally provide valuable information about their investment habits, including when they buy or sell stocks, what types of stocks they prefer, how frequently they trade, and how much risk they tend to take.

The key business problem of this project is therefore how to match quantitative investment strategies with client trading profiles. If a strategy has a trading style similar to that of a client and also achieves better risk-return performance than the client's own trading results, the client may be more willing to subscribe to and follow the strategy. Such a matching system can support more precise marketing and personalized investment advisory services. For instance, when a client logs into a trading application, the institution can recommend a strategy whose trading style, industry preference, and stock selection pattern are close to the client's historical behavior, while also showing the potential improvement in return and drawdown.

1.2 Research Questions

Based on the above background, this report aims to construct a data-driven and interpretable framework for matching securities investment strategies with client trading profiles. Specifically, we focus on the following three research questions.

First, can meaningful and interpretable features be extracted from transaction records to describe both strategies and clients? Since raw trading records are often detailed and irregular, it is necessary to summarize them into structured variables that reflect trading behavior, portfolio structure, and risk-return characteristics.

Second, can strategies and clients form stable and interpretable groups in a multivariate feature space? If clustering methods reveal clear groups, this suggests that different trading styles can be identified statistically rather than only described qualitatively.

Third, can we build a matching model that considers both style similarity

and performance improvement? In practice, a recommended strategy should not only look similar to the client’s own trading behavior, but also provide better investment outcomes in terms of return, risk, drawdown, or volatility.

1.3 Brief Literature Review

This project is related to several areas of multivariate statistical analysis and financial data mining. In investment analysis, portfolio characteristics and risk-return indicators are commonly used to describe the behavior and performance of investment strategies. Measures such as return, volatility, maximum drawdown, win rate, and Sharpe ratio help evaluate whether a strategy has attractive performance relative to its risk exposure.

In client profiling and recommendation problems, historical behavior data are often transformed into quantitative features before applying statistical or machine learning models. Similarity-based recommendation methods are especially useful when the objective is to find items or strategies that resemble a user’s historical preferences. In the context of this project, the “items” are quantitative trading strategies, while the “users” are clients with observable transaction records.

Multivariate statistical methods provide useful tools for this type of problem. Principal component analysis can reduce the dimensionality of the feature space and reveal the main sources of variation among clients and strategies. Cluster analysis, including K-means clustering, hierarchical clustering, and Gaussian mixture models, can be used to identify groups with similar trading profiles. Discriminant analysis can further examine whether the groups obtained from clustering are separable in a lower-dimensional projection. These methods are suitable for this project because they are interpretable and can directly connect statistical results with business meanings.

1.4 Structure of the Report

The remainder of this report is organized as follows. Section 2 introduces the feature engineering process and the statistical methods used in this study, including principal component analysis, clustering methods, discriminant analysis, and the two-stage client-strategy matching model. Section 3 describes the simulation study, where additional simulated clients are generated to enrich the sample space and test the feasibility of the proposed framework. Section 4 presents the empirical analysis, including data description, analysis procedures, clustering results, matching results, and verification of the results. Section 5 summarizes the main visualizations used to support the analysis. Section 6 concludes the report by discussing the main findings, limitations, and possible directions for future improvement.

2 Description of the Methods and Models

This section describes the main statistical methods and models used in this report. Since the objective of this project is to match clients with suitable quantitative investment strategies, the methodology is designed to be both interpretable and practical. The overall workflow includes feature engineering, data standardization, principal component analysis, cluster analysis, discriminant analysis, a two-stage client-strategy matching model, and statistical verification.

2.1 Feature Engineering

The original transaction records contain detailed information about trading time, stock code, trading direction, transaction volume, transaction amount, and strategy or client identity. However, raw transaction records cannot be directly used for multivariate statistical analysis. Therefore, we first transform the raw trading records into structured numerical features.

The final feature system is constructed from three perspectives: trading behavior, portfolio structure, and risk-return performance.

First, trading behavior features are used to describe how actively and regularly a strategy or client trades. These features include trading frequency, average holding period, turnover rate, and average transaction amount. Since trading frequency and average transaction amount may contain extreme values, logarithmic transformations are applied to reduce the influence of very active accounts or very large transactions. For example, if x denotes the original trading frequency or transaction amount, the transformed variable is defined as

$$\log(1 + x).$$

Second, portfolio structure features are used to describe the concentration and diversification of holdings. These features include buy-sell ratio, trading regularity, number of traded stocks, top-one and top-three concentration ratios, average position ratio, position peak ratio, and stock turnover. In particular, the position peak ratio is used to measure the difference between the maximum position and the average position, which helps reduce redundancy caused by using highly related position variables at the same time.

Third, risk-return features are used to evaluate investment performance. These features include average return, win rate, profit-loss ratio, maximum drawdown, return volatility, and approximate Sharpe ratio. For strategy data, return and drawdown information can be obtained from strategy performance records. For client accounts, because complete daily asset values and initial holding costs are not fully available, realized return and drawdown are approximately estimated based on matched buy and sell transactions.

2.2 Data Standardization

Before applying multivariate statistical methods, all numerical features are standardized. Standardization is necessary because different variables have different units and scales. For example, transaction amount may be much larger in magnitude than win rate or drawdown. Without standardization, variables with larger scales may dominate distance-based methods such as K-means clustering.

Let x_{ij} be the value of feature j for entity i . The standardized value z_{ij} is calculated as

$$z_{ij} = \frac{x_{ij} - \bar{x}_j}{s_j},$$

where $\bar{x}_j$ is the sample mean of feature j , and s_j is the sample standard deviation of feature j . All subsequent multivariate analyses are based on the standardized feature matrix.

2.3 Principal Component Analysis

Principal component analysis is used to explore the main sources of variation in the high-dimensional feature space. Since the feature system contains multiple correlated variables, PCA helps reduce dimensionality and provides a low-dimensional representation of clients and strategies.

Suppose the standardized data matrix is denoted by Z . PCA finds linear combinations of the original standardized variables:

$$PC_k = za_k,$$

where a_k is the loading vector of the k -th principal component. The loading vectors are obtained by solving the eigenvalue problem:

$$Sa_k = \lambda_k a_k,$$

where S is the covariance matrix of the standardized data, and λ_k is the eigenvalue corresponding to the k -th principal component.

The explained variance ratio of the k -th principal component is given by

$$\frac{\lambda_k}{\sum_{j=1}^p \lambda_j},$$

where p is the number of original features. In this report, PCA is mainly used for two purposes. First, it helps examine whether the extracted features contain strong common structures. Second, it provides two-dimensional visualizations that help compare strategies and clients in a lower-dimensional space.

2.4 Cluster Analysis

Cluster analysis is used to examine whether clients and strategies can be divided into several groups according to their trading profiles. If entities within the

same group have similar trading behavior, portfolio structure, and risk-return characteristics, the clustering results can provide useful information for client segmentation and strategy classification.

This report compares three clustering methods: K-means clustering, hierarchical clustering, and Gaussian mixture models.

2.4.1 K-means Clustering

K-means clustering partitions all entities into K clusters by minimizing the within-cluster sum of squared distances. The objective function is

$$\min_{C_1, \dots, C_K} \sum_{k=1}^K \sum_{i \in C_k} \|z_i - \mu_k\|^2,$$

where z_i is the standardized feature vector of entity i , C_k is the k -th cluster, and μ_k is the centroid of cluster C_k .

K-means is simple and interpretable, so it is used as the main clustering method in this report. Different values of K are compared, and the final number of clusters is selected according to clustering quality and interpretability.

2.4.2 Hierarchical Clustering

Hierarchical clustering is used as a robustness check for K-means clustering. In this report, Ward's method is applied. Ward's method merges clusters by minimizing the increase in within-cluster variance. Compared with K-means, hierarchical clustering does not require random initialization and can provide a dendrogram to show the nested structure among entities.

2.4.3 Gaussian Mixture Model

The Gaussian mixture model assumes that the data are generated from a mixture of several multivariate normal distributions. Its probability density function is

$$f(z_i) = \sum_{k=1}^K \pi_k \phi(z_i \mid \mu_k, \Sigma_k),$$

where π_k is the mixing weight of component k , μ_k is the mean vector, Σ_k is the covariance matrix, and $\phi(\cdot)$ is the multivariate normal density function.

Compared with K-means, GMM provides a probabilistic clustering result. Each entity has a probability of belonging to each cluster. This is useful because some clients or strategies may lie between different trading styles.

2.5 Evaluation of Clustering Quality

To evaluate the quality and stability of clustering results, several indicators are used.

The silhouette coefficient measures how similar an entity is to its own cluster compared with other clusters. A larger silhouette coefficient indicates better cluster separation.

The Davies-Bouldin index measures the average similarity between each cluster and its most similar cluster. A smaller Davies-Bouldin index usually indicates better clustering quality.

The adjusted Rand index is used to compare the consistency between different clustering methods. If two methods produce similar cluster assignments, the adjusted Rand index will be close to 1. In this report, ARI is used to compare K-means clustering with hierarchical clustering and GMM.

2.6 Linear Discriminant Analysis

After obtaining clustering labels, Fisher’s linear discriminant analysis is used to examine whether the clusters can be separated in a lower-dimensional discriminant space. LDA seeks linear combinations of features that maximize the between-cluster variation and minimize the within-cluster variation.

For a projection vector w , the Fisher criterion is

$$J(w) = \frac{w^T S_B w}{w^T S_W w},$$

where S_B is the between-class scatter matrix and S_W is the within-class scatter matrix.

In this report, LDA is not used as the main classification model. Instead, it is used as a visualization and verification tool. If the clusters obtained from unsupervised learning can be clearly separated in the LDA projection, this provides additional evidence that the clustering structure has statistical meaning.

2.7 Two-stage Client-Strategy Matching Model

The final objective of this project is to recommend suitable strategies to clients. Therefore, we construct a two-stage matching model that combines style similarity and performance improvement.

The first stage evaluates style similarity. For each client-strategy pair, we calculate the similarity between their trading behavior features and portfolio structure features. Cosine similarity is used because it measures the angle between two feature vectors and is less affected by the absolute magnitude of the vectors. For two standardized feature vectors u and v , cosine similarity is defined as

$$\text{sim}(u, v) = \frac{u^T v}{\|u\| \|v\|}.$$

The overall style score is calculated as a weighted average of behavior similarity and holding-structure similarity:

$$\text{style_score} = 0.45 \times \text{behavior_similarity} + 0.55 \times \text{holding_similarity}.$$

The second stage evaluates performance improvement. A recommended strategy should not only be similar to the client, but also perform better than the client’s own trading results. Therefore, return, Sharpe ratio, maximum drawdown, and return volatility are considered. Higher return and Sharpe ratio are preferred, while lower drawdown and volatility are preferred.

The final matching score is defined as

$$\text{final_score} = 0.70 \times \text{style_score} + 0.30 \times \text{performance_score}.$$

This design reflects the main logic of the project. Style similarity ensures that the recommended strategy is close to the client’s historical trading habits, while the performance score ensures that the strategy has the potential to improve the client’s investment outcome.

2.8 Kruskal-Wallis Test

Finally, the Kruskal-Wallis test is used to examine whether different clusters have statistically different feature distributions. Since the sample size is limited and the normality assumption may not hold, the Kruskal-Wallis test is more appropriate than one-way ANOVA for this project.

For each selected feature, the null hypothesis is that the distributions of this feature are the same across different clusters. A small p -value suggests that at least one cluster differs from the others. In this report, the test is used to check whether the clusters are related to meaningful variables such as win rate, position peak ratio, return volatility, and number of stocks.

Overall, the methods described above form a complete analytical framework. Feature engineering transforms raw transaction records into interpretable variables. PCA explores the main structure of the data. Clustering methods identify groups of similar clients and strategies. LDA provides additional visualization and verification. The two-stage matching model produces strategy recommendations by combining similarity and performance. Finally, statistical tests are used to examine whether the obtained groups have meaningful differences.

3 Simulation Studies

The original client sample contains only three simulated client accounts, which is not sufficient to support stable cluster analysis, discriminant analysis, or client segmentation. With such a small number of clients, the empirical results may be highly sensitive to individual accounts and may not fully reflect the diversity of possible client trading profiles. Therefore, this report introduces a simulation study to expand the client sample space and to examine whether the proposed analytical framework can still work under more heterogeneous client profiles.

It should be emphasized that the simulated clients are not intended to replace real business evidence. Instead, they are used as an auxiliary tool to test the feasibility and robustness of the proposed workflow. The main purpose of simulation is to create additional client profiles with different trading styles,

so that methods such as principal component analysis, clustering, discriminant analysis, and the client-strategy matching model can be evaluated in a richer multivariate setting.

3.1 Motivation for Simulation

The project aims to match quantitative investment strategies with client trading profiles. In a real securities institution, a large number of client accounts would usually be available, and these accounts would naturally contain diverse trading behaviors, holding structures, and risk preferences. However, in this project, only three original client accounts are provided. If the analysis is conducted only on these three clients, the sample size would be too limited to represent different types of investors.

To address this limitation, we generate 25 additional simulated clients based on the observed characteristics of the original client accounts and reasonable business assumptions. After adding the simulated clients, the final dataset contains both strategy entities and client entities, making it possible to conduct multivariate analysis in a more stable and interpretable way.

3.2 Design of Simulated Client Profiles

The simulated clients are generated according to three sources of information. First, we use the empirical mean and standard deviation of the original client features as basic references. Second, we incorporate business knowledge about common investor types in securities trading. Third, we impose reasonable financial boundaries on key variables to avoid unrealistic simulated values.

In this report, the simulated clients are divided into five representative types:

Table 1: Types of simulated clients

Client Type	Main Characteristics
Conservative defensive type	Trades less frequently, prefers relatively stable holdings, has lower return volatility, and usually accepts lower risk exposure.
Balanced allocation type	Holds a moderately diversified portfolio, has medium trading frequency, and seeks a balance between return and risk.
High-frequency trading type	Trades frequently, has higher turnover rate and shorter holding period, and may show larger transaction activity.
Growth aggressive type	Prefers growth-oriented stocks or sectors, accepts higher volatility, and aims for higher potential returns.
Concentrated holding type	Holds a small number of stocks with relatively high portfolio concentration, and may have higher idiosyncratic risk.

These five types are designed to cover common differences in client trading behavior, including trading activity, holding concentration, diversification level, and risk preference. They also help the model distinguish clients with different investment habits in the same feature space as the quantitative strategies.

3.3 Generation Procedure

The simulation procedure follows the feature system constructed in Section 2. For each simulated client, values are generated for trading behavior features, portfolio structure features, and risk-return features.

For trading behavior features, variables such as trading frequency, average holding period, turnover rate, and average transaction amount are generated around the observed client-level statistics. For example, high-frequency clients are assigned relatively higher trading frequency and turnover rate, while conservative clients are assigned lower trading frequency and longer holding periods.

For portfolio structure features, variables such as number of stocks, top-one concentration, top-three concentration, average position ratio, and position peak ratio are adjusted according to the assumed client type. For instance, concentrated holding clients are assigned higher concentration ratios and fewer stocks, while balanced allocation clients are assigned more diversified holdings.

For risk-return features, variables such as average return, win rate, profit-loss ratio, maximum drawdown, return volatility, and approximate Sharpe ratio are generated within reasonable financial ranges. Growth aggressive clients may have higher expected return but also higher volatility and drawdown. Conservative defensive clients usually have lower volatility and smaller drawdown, but their expected return may also be lower.

3.4 Boundary Control

To ensure that the simulated clients are financially meaningful, boundary controls are applied during the generation process. Variables such as win rate, position ratio, concentration ratio, and drawdown must remain within reasonable ranges. For example, win rate should be between 0 and 1, portfolio concentration should not exceed 1, and volatility and drawdown should not take impossible negative values.

This boundary control is important because unrealistic simulated observations may distort PCA directions, clustering results, and similarity scores. Therefore, the simulation process is not a purely random data-generation exercise. It is guided by both the original client data and the financial meaning of each variable.

3.5 Role of Simulation in the Analysis

The simulated clients play three main roles in this report.

First, they enlarge the client sample and make the multivariate analysis more stable. With more client observations, PCA and clustering can better reveal the structure of the feature space.

Second, they provide a richer set of client profiles. This allows us to check whether the proposed matching model can handle clients with different trading styles, rather than only the three original accounts.

Third, they help evaluate the feasibility of the whole workflow. If the feature engineering, clustering, discriminant analysis, and matching model can operate reasonably on the expanded dataset, then the framework can be considered a feasible preliminary solution for the client-strategy matching problem.

However, the simulation results should be interpreted carefully. The simulated clients do not represent the real distribution of all investors in the securities market. They are mainly used for methodological testing and exploratory analysis. Therefore, the final business conclusion should still rely primarily on real client data when more complete account records become available.

4 Empirical Analysis and Visualization

4.1 Data Acquisition and Data Description

The empirical analysis is conducted on both quantitative strategy data and client account data. According to the project setting, the main task is to compare the transaction records of quantitative strategies with the historical trading records of clients, and then recommend strategies that are similar to the clients' trading profiles and have better risk-return performance. The original project data include strategy performance files and simulated client account files. In the final integrated dataset used in this report, we include 20 quantitative strategies, 3 real simulated client accounts, and 25 additional simulated clients. Therefore, the final empirical dataset contains 48 analytical entities in total.

The strategy data are obtained from strategy performance Excel files. These strategies cover several investment themes, including large-cap enhancement, small- and mid-cap enhancement, technology sectors, consumption defense, cyclical rotation, momentum trading, balanced allocation, and other quantitative styles. The client data are obtained from three simulated account files. These accounts provide transaction-level information that can be used to describe each client's trading behavior and holding structure.

For each strategy or client, we construct an 18-dimensional feature vector. The variables are divided into three groups: trading behavior features, portfolio structure features, and risk-return features. Trading behavior features describe how frequently and actively an entity trades. Portfolio structure features describe diversification, concentration, and position patterns. Risk-return features describe investment performance, volatility, drawdown, and approximate risk-adjusted return. Before empirical analysis, all features are standardized so that variables with different scales can be compared in the same multivariate space. The correlation structure among these features is shown in Figure 1.

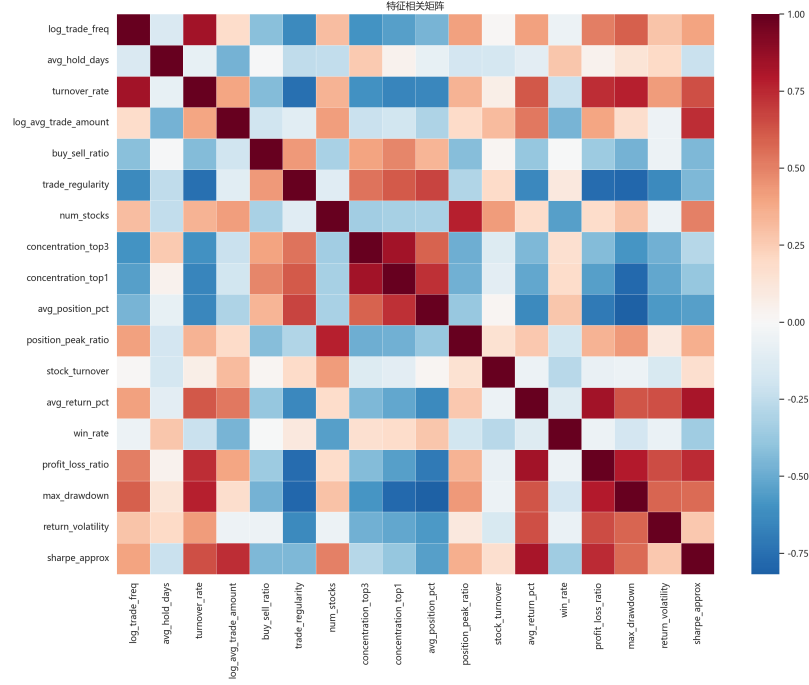


Figure 1: Correlation matrix of the 18-dimensional feature system

4.2 Analysis Procedure

The empirical analysis follows the methodological framework described in Section 2. First, principal component analysis is applied to the standardized feature matrix to explore the main variation directions of the data. Second, K-means clustering, hierarchical clustering, and Gaussian mixture models are used to identify groups of clients and strategies with similar trading profiles. Third, Fisher's linear discriminant analysis is used to visualize whether the clusters obtained from K-means can be separated in a discriminant space. Fourth, the two-stage matching model is applied to the three real simulated clients in order to identify their most suitable strategies. Finally, clustering quality and statistical significance are examined through validation indicators and non-parametric tests.

4.3 Results of Principal Component Analysis

Principal component analysis is used to reduce the dimensionality of the 18-dimensional feature space and to summarize the main sources of variation among strategies and clients. The first five principal components explain a large proportion of the total variance, which suggests that the original features contain a strong common structure.

Table 2: Explained variance ratios of the first five principal components

Principal Component	Explained Variance Ratio	Cumulative Explained Variance
PC1	44.41%	44.41%
PC2	15.23%	59.64%
PC3	8.78%	68.42%
PC4	6.19%	74.61%
PC5	5.77%	80.38%

As shown in Table 2, the first principal component explains 44.41% of the total variance, while the first two principal components explain 59.64% in total. The cumulative explained variance of the first five components reaches 80.38%. This result indicates that the high-dimensional feature system can be effectively summarized by a smaller number of principal components.

The PCA visualization results are presented in Figures 2 and 3. Figure 2 shows both the cumulative explained variance and the two-dimensional PCA projection. The first few principal components explain most of the total variance, indicating that the 18-dimensional feature system can be effectively summarized by a lower-dimensional structure. In the two-dimensional projection, real clients and simulated clients are mainly located on the left side, while most strategies are distributed on the right side. This suggests that strategies and clients have systematic differences in trading behavior, portfolio structure, and risk-return characteristics.

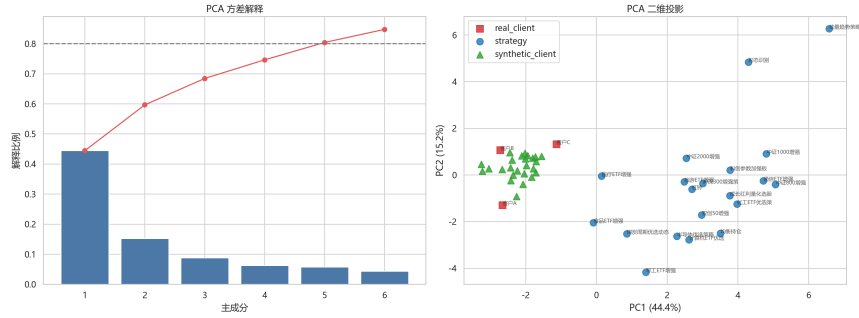


Figure 2: PCA explained variance and two-dimensional projection

Figure 3 shows the PCA loading plot of the original features. The directions and lengths of the arrows indicate how strongly different variables contribute to the first two principal components. Variables related to trading frequency, turnover rate, return, drawdown, and Sharpe ratio contribute strongly to the positive direction of PC1, while variables such as portfolio concentration and trading regularity contribute in different directions. This result helps explain the financial meaning of the principal components and supports the interpretability

of the PCA analysis.

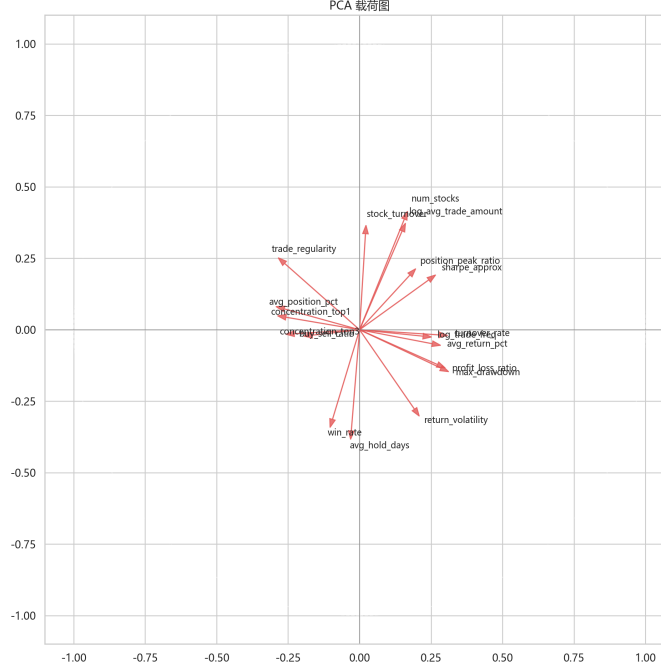


Figure 3: PCA loading plot of the 18-dimensional feature system

From the interpretation of the principal component loadings, PC1 mainly reflects the overall differences in trading scale, trading activity, and risk-return performance between strategies and clients. PC2 further captures differences related to return volatility and style differentiation. Therefore, PCA provides an intuitive low-dimensional representation for comparing different entities in the dataset.

4.4 Clustering Results

Cluster analysis is used to examine whether strategies and clients can form stable and interpretable groups in the standardized feature space. In the K-means clustering procedure, the candidate number of clusters is set from 2 to 6. For each value of K , we calculate the inertia, the silhouette coefficient, and the Davies–Bouldin index to evaluate clustering quality.

Figure 4 presents the diagnostic results for selecting the number of clusters. The inertia decreases as K increases, which is a common pattern in K-means clustering. However, the silhouette coefficient reaches its highest value when $K = 3$, while the Davies–Bouldin index also remains at a relatively low level. Considering both statistical indicators and business interpretability, we select

$K = 3$ as the final number of clusters.

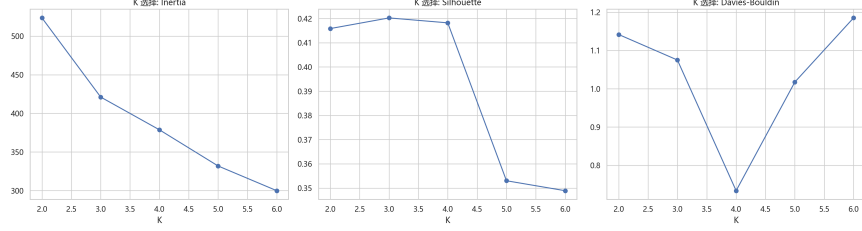


Figure 4: Diagnostic plots for selecting the number of clusters

Table 3: Clustering evaluation for different values of K

K	Silhouette Coefficient	Davies–Bouldin Index
2	0.4159	1.1418
3	0.4203	1.0752
4	0.4182	0.7337
5	0.3530	1.0175
6	0.3489	1.1852

Table 3 shows that $K = 3$ achieves the highest silhouette coefficient among the candidate values. Although $K = 4$ has a lower Davies–Bouldin index, the three-cluster solution is more interpretable and more consistent with the objective of distinguishing broad trading styles. Therefore, this report adopts $K = 3$ as the final number of clusters for K-means clustering.

To further examine the clustering structure, hierarchical clustering with Ward’s linkage is also applied. Figure 5 shows the dendrogram obtained from hierarchical clustering. The dendrogram provides a visual representation of how clients and strategies are gradually merged into larger groups. It can be observed that client entities and strategy entities tend to form distinguishable branches, which supports the existence of meaningful group structures in the feature space.

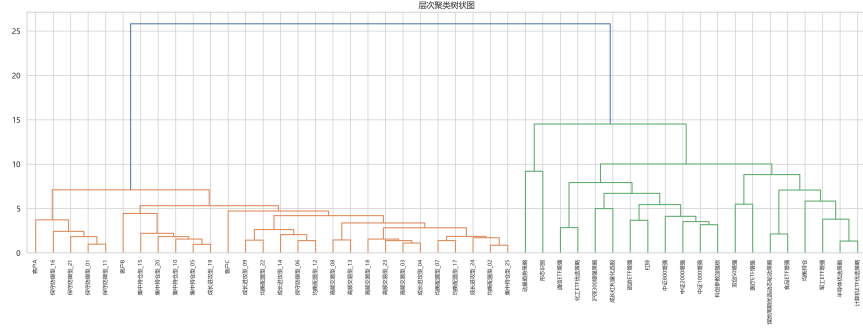


Figure 5: Dendrogram of hierarchical clustering

To evaluate the stability of the clustering results, we compare K-means clustering with hierarchical clustering and Gaussian mixture models. The adjusted Rand index between K-means and hierarchical clustering is 0.8438, while the adjusted Rand index between K-means and the Gaussian mixture model is 0.9699. These values are close to 1, indicating that the clustering results are highly consistent across different methods.

Figure 6 further illustrates the agreement among the three clustering methods. The high ARI values suggest that the discovered cluster structure is not merely a result of one specific algorithm. Instead, it reflects relatively stable patterns in the trading-profile feature space.

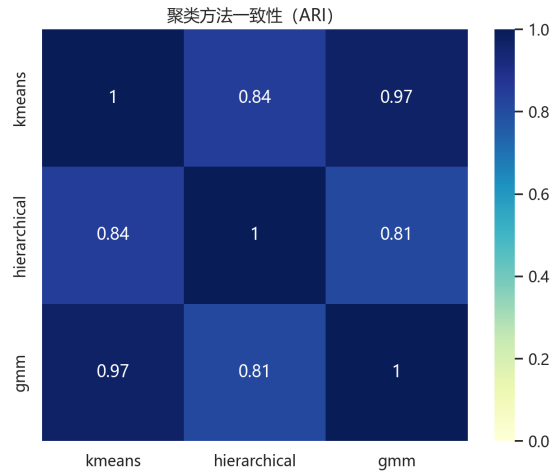


Figure 6: Agreement among different clustering methods measured by adjusted Rand index

4.5 Results of Linear Discriminant Analysis

After obtaining the K-means cluster labels, Fisher’s linear discriminant analysis is used to project the entities into a low-dimensional discriminant space. The purpose of this step is not to replace the unsupervised clustering methods, but to check whether the clusters can be separated using linear combinations of the original features.

The LDA projection shows that the strategy group and the client group can be distinguished to a certain extent in the discriminant space. Real clients and simulated clients are mainly located on the client side of the projection, while many strategies are located in regions with stronger risk-return and style characteristics. This result suggests that the difference between clients and strategies is not only caused by random noise, but is also reflected in systematic feature patterns.

Figure 7 presents the two-dimensional projection obtained from Fisher’s linear discriminant analysis. The horizontal axis represents the first discriminant direction, while the vertical axis represents the second discriminant direction. Different colors indicate different clusters, and different markers represent different entity types, including strategies, real clients, and simulated clients.

From the figure, most client-related entities are concentrated in the upper-right region of the discriminant space, while many strategies are distributed in the lower-right region. In addition, a small number of strategies are located far away from the main groups, suggesting that they have more distinctive trading or risk-return characteristics. This result indicates that the clusters obtained from the previous unsupervised analysis can also be separated in a supervised discriminant projection, which provides additional support for the validity of the grouping structure.

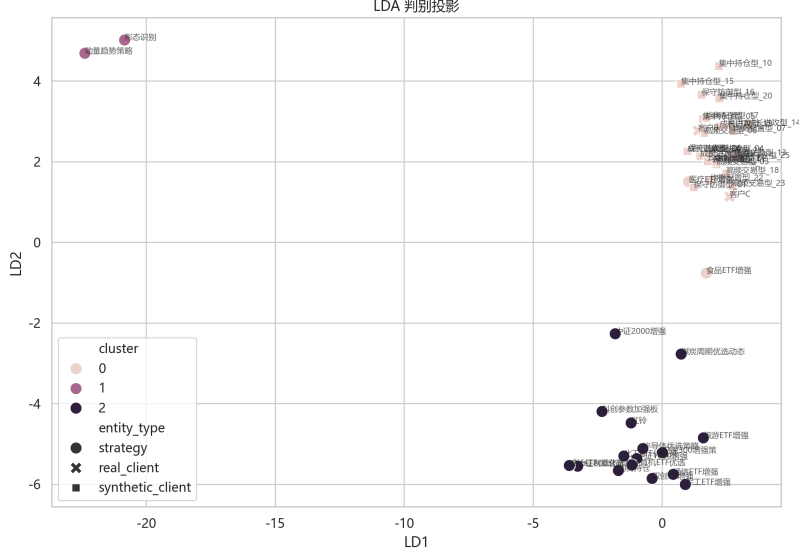


Figure 7: Two-dimensional projection of Fisher's linear discriminant analysis

The LDA result also provides an important implication for strategy recommendation. If we only consider style similarity, some strategies may appear close to a client but may not necessarily improve the client's performance. Therefore, it is necessary to combine style similarity with risk-return evaluation in the final matching model.

4.6 Client-Strategy Matching Results

The two-stage matching model is applied to the three real simulated clients. In the first stage, the model calculates style similarity based on trading behavior and portfolio structure. In the second stage, the model evaluates whether the strategy has better return, Sharpe ratio, drawdown, and volatility than the client's approximate historical performance. The final score is a weighted combination of the style score and the performance score.

Figure 8 presents the matching scores between the three real clients and all candidate strategies. A higher score indicates that the corresponding strategy is more suitable for the client after considering both trading-style similarity and risk-return performance. The heatmap provides an overall view of the recommendation pattern. It shows that different clients have clearly different high-scoring strategies, which means that the matching model does not simply recommend the same strategy to all clients.

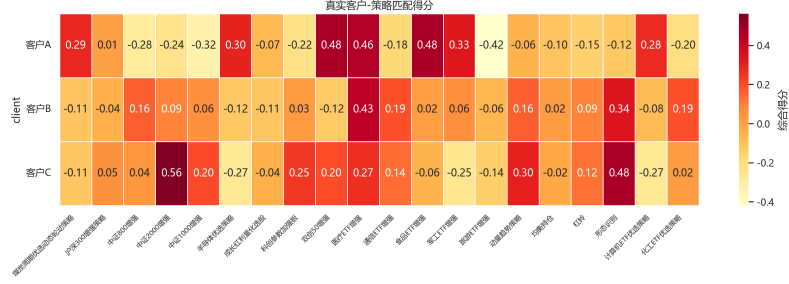


Figure 8: Matching score heatmap between real clients and quantitative strategies

Based on the final matching score, the top recommended strategy for each real simulated client is summarized in Table 4.

Client	Top Recommended Strategy	Final Score	Return Improvement	Drawdown Improvement
Client A	Shuangchuang 50 Enhancement	0.4810	1.1015	0.2580
Client B	Medical ETF Enhancement	0.4341	0.2115	0.8010
Client C	CSI 2000 Enhancement	0.5620	0.6706	0.3372

Figure 9 further shows the Top 5 recommended strategies for each real client. For Client A, the highest-scoring strategies are mainly enhancement or ETF-related strategies, which is consistent with the client’s relatively concentrated and less frequent trading profile. For Client B, the recommended strategies include Medical ETF Enhancement and several other diversified strategies, reflecting the client’s more active and diversified trading behavior. For Client C, the highest score is obtained by CSI 2000 Enhancement, which is consistent with the client’s growth-oriented and small- or mid-cap style exposure.

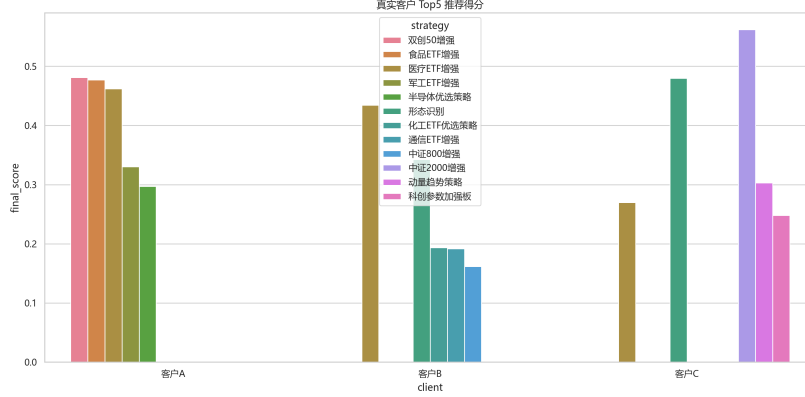


Figure 9: Top 5 recommended strategies for each real client

In addition to the final matching score, it is important to examine whether the recommended strategies actually improve the clients' investment performance. Figure 10 compares the return improvement and drawdown improvement of the Top 5 recommended strategies for the three real clients. The left panel shows the difference between strategy return and client return, while the right panel shows the improvement in maximum drawdown. The results indicate that the recommended strategies generally provide positive improvements, especially for the Top 1 strategies listed in Table 4.

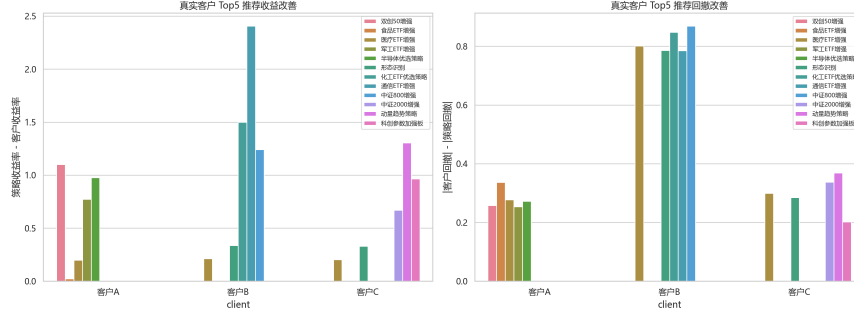


Figure 10: Return improvement and drawdown improvement of Top 5 recommendations

Figure 11 provides another perspective on the recommendation results. The horizontal axis represents the normalized final matching score, while the vertical axis represents the relative return improvement compared with the client's own performance. Points in the upper-right region are more desirable because they have both higher matching scores and larger return improvements. The real client recommendations are located above the zero-improvement line, which

supports the effectiveness of combining style similarity with performance evaluation.

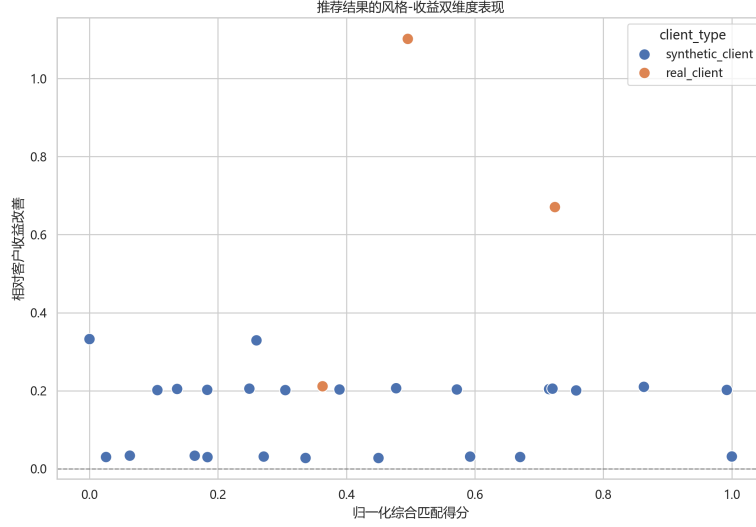


Figure 11: Style-performance trade-off of recommendation results

Overall, the matching results support the main idea of this project. The model recommends different strategies for different clients according to their trading profiles, and the top recommended strategies for the three real simulated clients all improve both return and maximum drawdown compared with the clients' approximate historical performance. Therefore, the proposed two-stage model can provide an interpretable and practical preliminary solution for client-strategy matching.

4.7 Verification of Results

To verify the empirical findings, this report considers both clustering consistency and statistical differences among clusters.

First, the adjusted Rand index is used to compare the clustering labels obtained from different methods. The ARI between K-means and hierarchical clustering is 0.8438, and the ARI between K-means and GMM is 0.9699. These results indicate that the clustering structure is stable across different clustering algorithms.

Second, the Kruskal-Wallis test is used to examine whether important variables differ significantly across clusters. Since the sample size is limited and the feature distributions may not satisfy the normality assumption, the Kruskal-Wallis test is more suitable than a standard one-way ANOVA.

Table 5 shows that several variables have statistically significant or marginally significant differences across clusters. In particular, win rate, position peak ra-

Table 5: Kruskal-Wallis test results for selected features

Feature	p -value
win_rate	0.0298
position_peak_ratio	0.0322
return_volatility	0.0324
num_stocks	0.0369
sharpe_approx	0.0894

tio, return volatility, and number of stocks have relatively small p -values. This suggests that the clusters are not formed purely by random variation. Instead, they are associated with meaningful financial and trading characteristics, such as winning probability, position structure, volatility level, and stock diversification.

Overall, the empirical results show that the proposed framework is feasible for the client-strategy matching problem. PCA reveals the main structure of the feature space, clustering methods identify stable groups, LDA provides visual evidence of group separation, and the two-stage matching model generates interpretable recommendations for the three real simulated clients. However, because the number of real client accounts is still limited and part of the client sample is simulated, the empirical conclusions should be interpreted as exploratory results rather than final business recommendations.

5 Conclusion

This report studies the problem of matching securities investment strategies with client trading profiles. Motivated by the need for personalized strategy recommendation in securities institutions, we construct an interpretable multivariate statistical framework based on transaction-level strategy data and client account data. The proposed framework transforms raw trading records into structured features, analyzes the main patterns in the feature space, identifies groups of similar entities, and finally recommends suitable strategies to clients by jointly considering style similarity and risk-return improvement.

The main contribution of this report is the construction of a complete client-strategy matching workflow. First, an 18-dimensional feature system is developed from three perspectives: trading behavior, portfolio structure, and risk-return performance. These features describe not only how frequently and actively an entity trades, but also how concentrated its portfolio is and how well it performs under risk. Second, principal component analysis is used to summarize the main variation directions in the feature space. The PCA results show that the original variables contain a strong common structure, and that clients and strategies can be compared in a lower-dimensional representation. Third, K-means clustering, hierarchical clustering, and Gaussian mixture models are applied to examine the grouping structure of clients and strategies. The con-

sistency among different clustering methods suggests that the identified groups are relatively stable. Fourth, Fisher’s linear discriminant analysis provides further visual evidence that the clusters can be separated in a discriminant space. Finally, a two-stage matching model is designed to recommend strategies by combining trading-style similarity with performance improvement.

The empirical results support the feasibility of the proposed framework. The final dataset contains 20 quantitative strategies, 3 real simulated client accounts, and 25 additional simulated clients. The K-means clustering result suggests that three clusters provide a reasonable and interpretable grouping structure. The adjusted Rand index between K-means and hierarchical clustering is 0.8438, while that between K-means and the Gaussian mixture model is 0.9699, indicating that the clustering structure is stable across different methods. In addition, the Kruskal–Wallis test shows that several important variables, such as win rate, position peak ratio, return volatility, and number of stocks, differ across clusters. This suggests that the clusters are related to meaningful financial and trading characteristics rather than random variation.

For the three real simulated clients, the matching results are also consistent with the project objective. Client A is recommended Shuangchuang 50 Enhancement, Client B is recommended Medical ETF Enhancement, and Client C is recommended CSI 2000 Enhancement. These Top 1 recommended strategies all improve both return and maximum drawdown compared with the clients’ approximate historical performance. Therefore, the results support the central idea of this project: transaction data can be used to identify client trading profiles and to recommend strategies that are both similar to the clients’ own trading styles and superior in risk-return performance.

However, the conclusions of this report should be interpreted as exploratory results rather than direct business recommendations. There are several limitations. First, the number of real client accounts is very limited. Although simulated clients are generated to expand the sample space and test the feasibility of the method, they cannot fully represent the true distribution of real investors. Second, the client-side performance indicators are approximate because complete initial holding costs, daily portfolio snapshots, and full account equity curves are not available. As a result, client return, drawdown, and Sharpe ratio are estimated from matched buy and sell records, which may differ from strict account-level performance measures. Third, the current feature system mainly focuses on transaction behavior, position structure, and risk-return characteristics. More detailed information, such as industry classification, market conditions, client risk questionnaires, and longer historical records, could further improve the recommendation accuracy.

Future work can improve this study in several directions. First, more real client accounts should be collected to validate whether the proposed framework remains stable in a larger and more realistic sample. Second, industry and sector preference features can be added to better capture clients’ stock selection habits. Third, complete daily asset curves can be used to calculate more accurate return, volatility, drawdown, and risk-adjusted performance indicators. Fourth, more advanced multivariate methods, such as MANOVA, Hotelling’s T^2

test, canonical correlation analysis, or supervised learning models, can be introduced to further examine the relationship between client profiles and strategy characteristics.

Overall, this report provides a feasible preliminary solution for precise matching between securities investment strategies and client trading profiles. Although the current analysis is limited by data availability, the results show that multivariate statistical methods can produce interpretable client profiles, stable grouping structures, and practical strategy recommendations. The proposed framework therefore has potential value for personalized investment advisory services and precision marketing in securities institutions.